Question ID: 928498f3

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear equations in one variable and systems of equations in two variables	Medium

Question

$6x^2+5x-7=0$

What are the solutions to the given equation?

Answer

A. $\dfrac{-5\pm\sqrt{25+168}}{12}$

B. $\dfrac{-6\pm\sqrt{25+168}}{12}$

C. $\dfrac{-5\pm\sqrt{36-168}}{12}$

D. $\dfrac{-6\pm\sqrt{36-168}}{12}$

Correct Answer: A

Rationale

Choice A is correct. The quadratic formula, $x=\dfrac{-b\pm\sqrt{b^2-4ac}}{2a}$, can be used to find the solutions to an equation in the form $ax^2+bx+c=0$. In the given equation, $a=6$, $b=5$, and $c=-7$. Substituting these values into the quadratic formula gives $\dfrac{-5\pm\sqrt{5^2-4(6)(-7)}}{2(6)}$, or $\dfrac{-5\pm\sqrt{25+168}}{12}$.

Choice B is incorrect and may result from using $\dfrac{-a\pm\sqrt{b^2-4ac}}{2a}$ as the quadratic formula. Choice C is incorrect and may result from using $\dfrac{-b\pm\sqrt{a^2+4ac}}{2a}$ as the quadratic formula. Choice D is incorrect and may result from using $\dfrac{-a\pm\sqrt{a^2+4ac}}{2a}$ as the quadratic formula.